

UNIBUS LIVE

Real-Time University Bus Tracking — Live Map, Arrival Times & Alerts, in English & Bangla

TEAM UNIBUS

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INTRODUCTION

UniBus Live shows every university bus **live on a map** with the **arrival time for each stop**. A GPS tracker fixed in the bus reports every 5 seconds; our server cleans the data and pushes position + ETAs to every student's phone **within seconds**.

OBJECTIVES

- See the bus & your stop's arrival time **before leaving home**
- No driver dependence** — the fixed tracker works by itself
- Affordable (one low-cost server) & bilingual: **English + Bangla**

PROBLEM STATEMENT

Students wait at stops with no idea whether the bus left five minutes ago or is fifteen minutes away — missed buses, long waits, crowding. Commercial trackers are costly and English-only.

WHAT WE BUILT & SOLVED

- ✓ Live map + **arrival time for every stop**, updated in seconds
- ✓ **Map never blank** — bus visible at the depot before departure
- ✓ Noisy GPS cleaned — impossible jumps & jitter **filtered out**
- ✓ **GPS tracker alone runs it** — tracks the bus automatically, no driver or phone needed
- ✓ Driver's phone is an **optional backup only** — steps in if a bus has no tracker or it goes offline
- ✓ **Get-off alert** when the bus is within 80 m of your stop

KEY HARDWARE & STACK

Concox GT06S GPS tracker
fixed in the bus · GPS + GSM · fully automatic

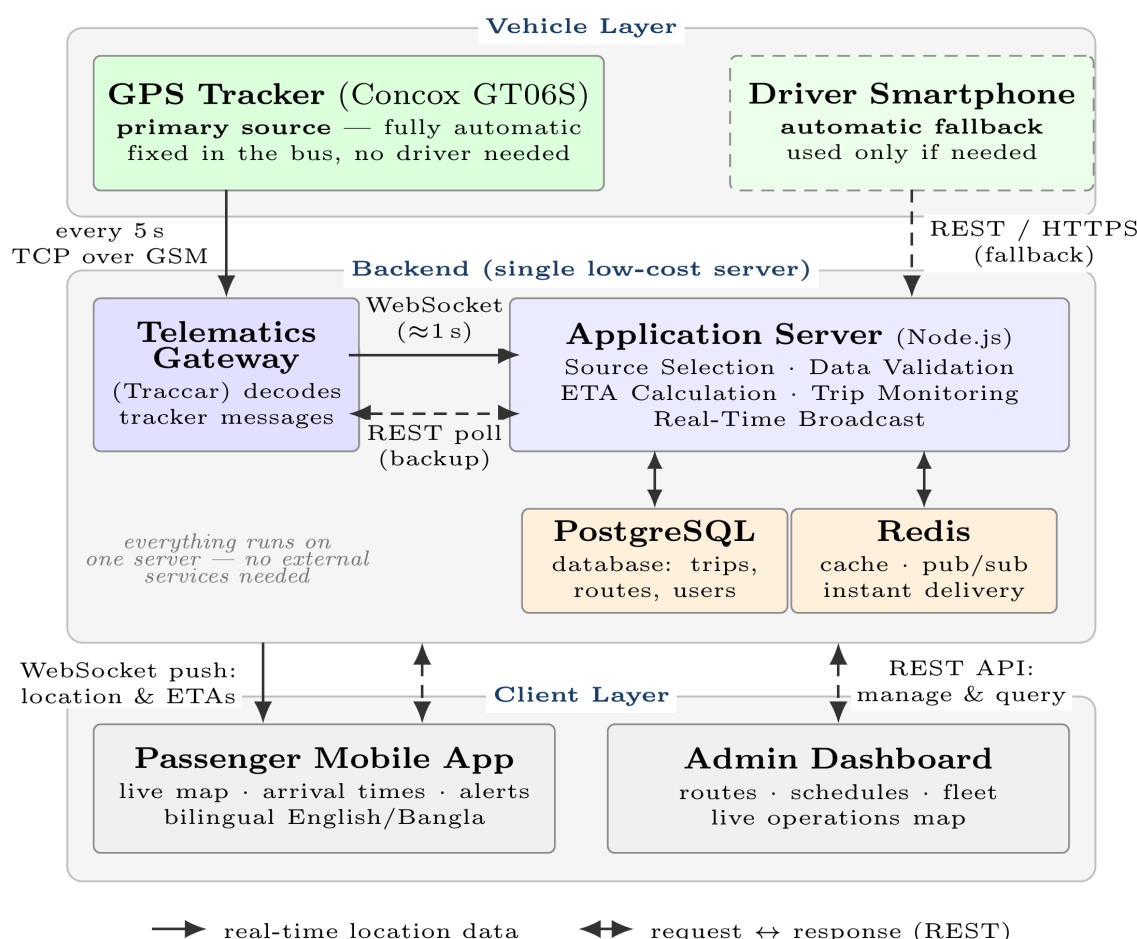
Driver smartphone
backup source only, used if the tracker is silent

Cloud server (VPS)
one low-cost machine runs everything

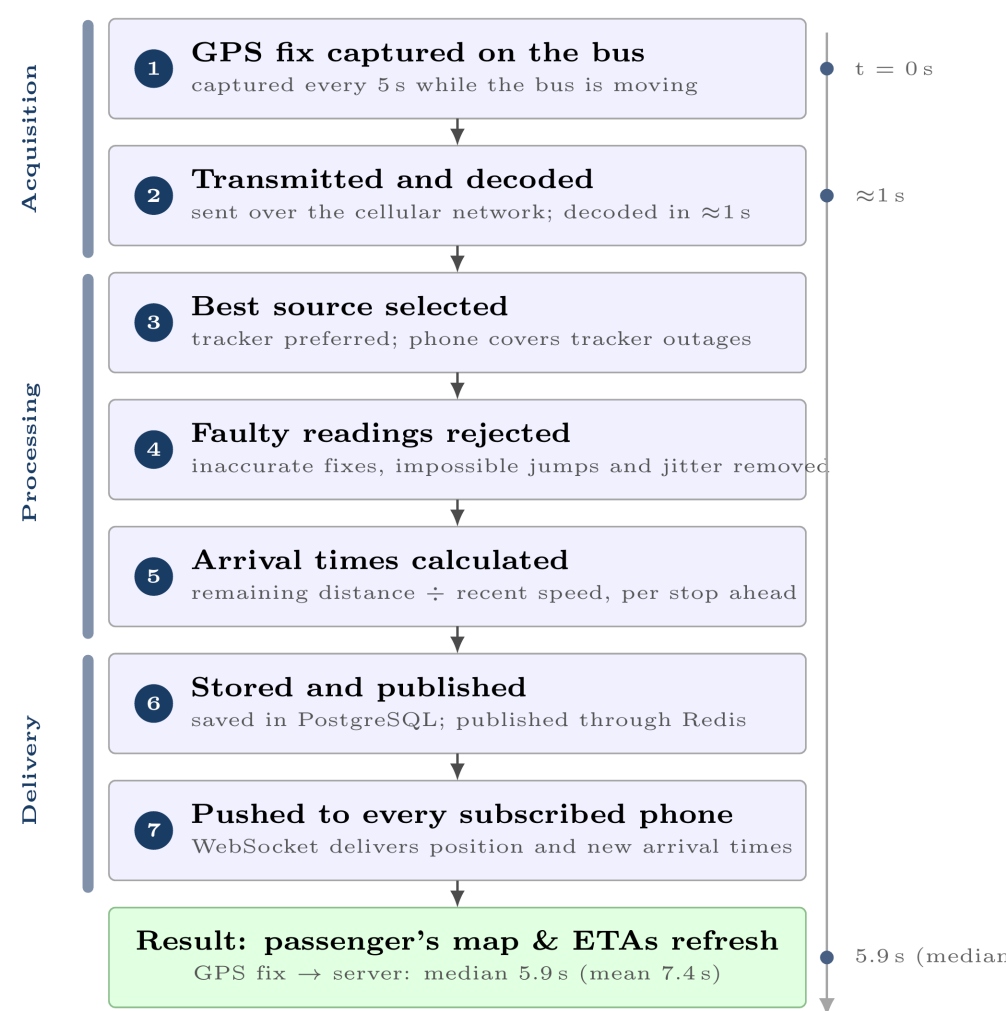
Passenger smartphones
live map + notifications for every rider

Node.js PostgreSQL Redis Traccar Socket.IO React Native Next.js

SYSTEM ARCHITECTURE & SETUP



METHODOLOGY



The journey of one position report — from the GPS fix on the bus to the rider's screen.

RESULTS & OUTPUT

5.9 s
median delay,
GPS fix → server

46,548
GPS reports
handled

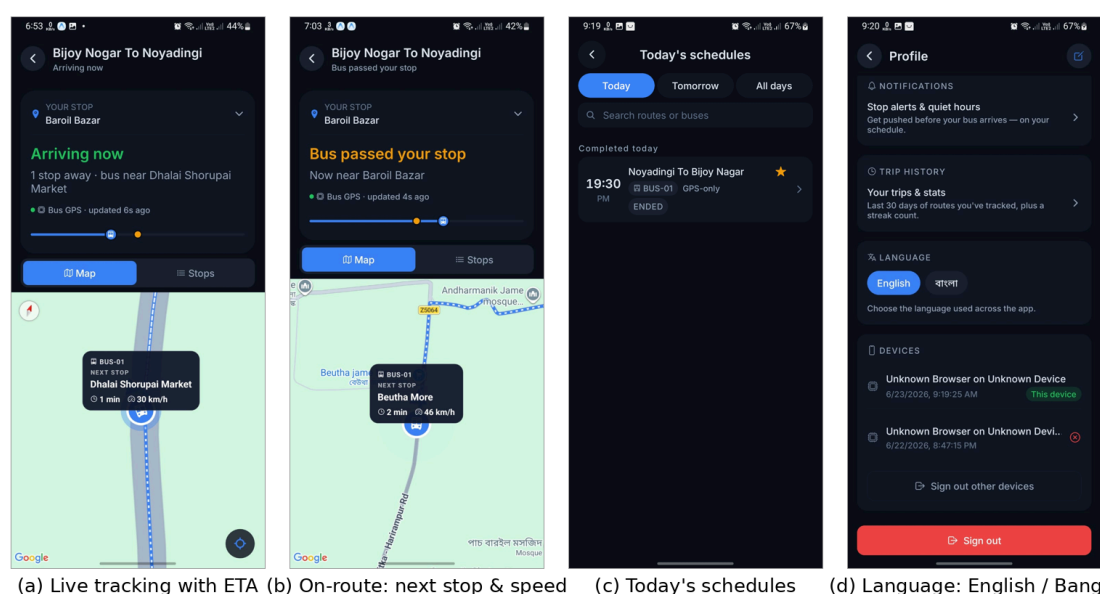
71
real trips tracked
across 9 routes

64 / 36
% trips: tracker vs phone
(only 1 of 3 buses fitted)

80%
arrival alerts
read by riders

80 m
get-off alert radius
at your stop

Proven in a 7-week pilot on the real university fleet — 3 buses · 9 routes · 38 riders · zero simulation



The bilingual passenger app: live tracking with ETA · on-route status · today's schedules · English/Bangla toggle

A HIDDEN PROBLEM — FOUND & FIXED

98%
of 46,548 GPS reports were pure **waste**
— buses kept transmitting every 5 s
even while parked overnight

60×
less parked-time reporting after our fix
— tracker **5 s → 300 s** when parked,
server polling **8 s → 60 s** when idle

FUTURE SCOPE

- Smarter ETAs** — an AI model trained on our logged trip history
- Trackers on every bus**, plus seat-occupancy (crowd) info
- Automatic on-time reports** for the university administration

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